

The Verified-Candidate Loop for Budgeted Large Language Model-Assisted High-Level Synthesis

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Abstract—High-level synthesis (HLS) agents must repair and optimize unfamiliar kernels under limited model and tool budgets. The Verified-Candidate Loop (VCL) retains a verified fallback, promotes a candidate only after seven validation gates, and ranks valid candidates with a bounded composite score. Deterministic diagnostic extraction and public-evidence retrieval constrain later model proposals. We evaluate three deployed agent–endpoint configurations on a frozen, balanced 150-task suite. Final completion is 143/150 for DeepSeek V4 Pro, 140/150 for Qwen3.5-122B-A10B, and 141/150 for Qwen3.6-27B. All 424 outputs that pass the public gates also pass our evaluator-side held-out gate suite, which is distinct from the organizer’s final hidden benchmarks. Qwen3.5 records the highest paired composite scores (77.77, 78.93) on 24 common tasks, while DeepSeek uses the fewest observed tokens.

Index Terms—high-level synthesis, LLM agent, FPGA, retrieval-augmented generation, quality of results

I. GATE EVIDENCE, NOT PROMPTS

LLM agents can edit unfamiliar HLS kernels, but prompts alone do not establish correctness or hardware quality. Tool reports provide reproducible, but test-bounded, evidence. A metered agent therefore needs a promotion rule that preserves a verified fallback whenever a required validation gate fails [1].

Prior HLS agents automate code transformation, directive search, and feedback-driven repair [2]–[4]. Tool-using language agents alternate reasoning and actions [5]. Our design assigns candidate promotion to deterministic HLS evidence under explicit model and tool budgets.

The Verified-Candidate Loop (VCL) presents three elements. **(1) Validation contract.** A fail-closed workflow preserves a verified fallback across Interface, CSIM, SYNTH, Frequency, Capacity, required CoSIM, and Metric-completeness gates. **(2) Composite promotion utility.** The bounded Q_{HW} combines measured performance and resource ratios with source-change and repair evidence; despite its implementation name, it is not a pure hardware QoR metric. **(3) Artifact-backed evaluation.** Completion uses the fixed 150-task denominator, while held-out-gate pass and paired QoR retain their applicable denominators; final records and aggregation scripts are released (Section IV).

II. TOOL EVIDENCE CONTROLS THE AGENT WORKFLOW

The task interface exposes an editable kernel, public tests, target part, and budgets while withholding evaluator-side tests and frozen references. The controller records every candidate, gate, metric, token, and credit in one run state.

The VCL runs seven gates in order: Interface, CSIM, SYNTH, Frequency, Capacity, required CoSIM, and Metric completeness. Frequency enforces 100 MHz, Capacity enforces U55C limits,

and CoSIM applies only when required. Define validity $V(c) = \prod_i G_i$ over applicable gates $G_i \in \{0, 1\}$; an inapplicable gate contributes 1. Let b be the verified fallback. The promotion rule is $P(c) = V(c) \wedge [Q_{\text{HW}}(c) > Q_{\text{HW}}(b)]$.

Local failure classification and retrieval require no additional model call; generating a subsequent candidate still does. **Failure Reflection** parses interface, synthesis, and co-simulation diagnostics plus the source diff into a compact constraint for the next proposal. **QoR-RAG** retrieves compatible public rules and measured success/failure cases by task metadata and bottleneck, then adds them to the optimization context. Thus interface mismatches produce signature/pragma constraints, synthesis failures produce timing/resource/II constraints, and co-simulation deadlocks produce handshake/port-width constraints. Duplicate candidates and repeated failures stop retries.

Figure 1 summarises the loop. The controller records every candidate and gate outcome for audit, while a failed or incomplete candidate never replaces the last verified fallback.

III. A COMPOSITE SCORE REJECTS COSTLY GAINS

Q_{HW} is the implementation’s bounded promotion utility, not a hardware-only objective. For anchor a and candidate c , let $r_L = (p_a L_a)/(p_c L_c)$ use cycle latency L and effective clock period p . When initiation interval applies, $r_P = r_L^{0.85} (II_a/II_c)^{0.15}$; otherwise $r_P = r_L$. Define the capacity-normalized resource footprint $A(x) = \sum_r x_r/K_r$ over LUT, FF, DSP, BRAM, and URAM, and $r_A = A(a)/A(c)$. Let $D \in \{0, 1\}$ mark a source change and $R \in \{0, 1\}$ mark a valid candidate that repairs an invalid starter. The composite ratio is $h = 0.99^D 2^R r_P^{0.55} r_A^{0.45}$.

$Q_{\text{HW}} = 1 - 1/(1 + h)^2$ maps $h=1$ to 0.75, approaches 1 for large gains, and approaches 0 for regressions. Budget efficiency $E = \max(0.80, 1 - 0.10u_{\text{credit}} - 0.10u_{\text{time}})$ uses consumed fractions clipped to $[0, 1]$. The final score is

$$S = 100 V(c) Q_{\text{HW}} E. \quad (1)$$

The 0.99 and 2 factors encode anti-no-op and repair policies, while 0.85/0.15 is a fixed latency/II split. The 0.55/0.45 performance/resource balance was heuristically calibrated against 95 official PPA references using 99 fresh Vitis 2025.2 measurements; 0.55 lies inside their feasible interval. This supports a common operating point, but the evidence is limited and one fixed exponent need not fit every task. Promotion occurs only after $V(c)=1$, and a failed gate yields $S=0$.

IV. EVALUATION ON A FROZEN TASK SUITE

Protocol. Each configuration processes the same frozen, balanced 150-task suite constructed from four public HLS

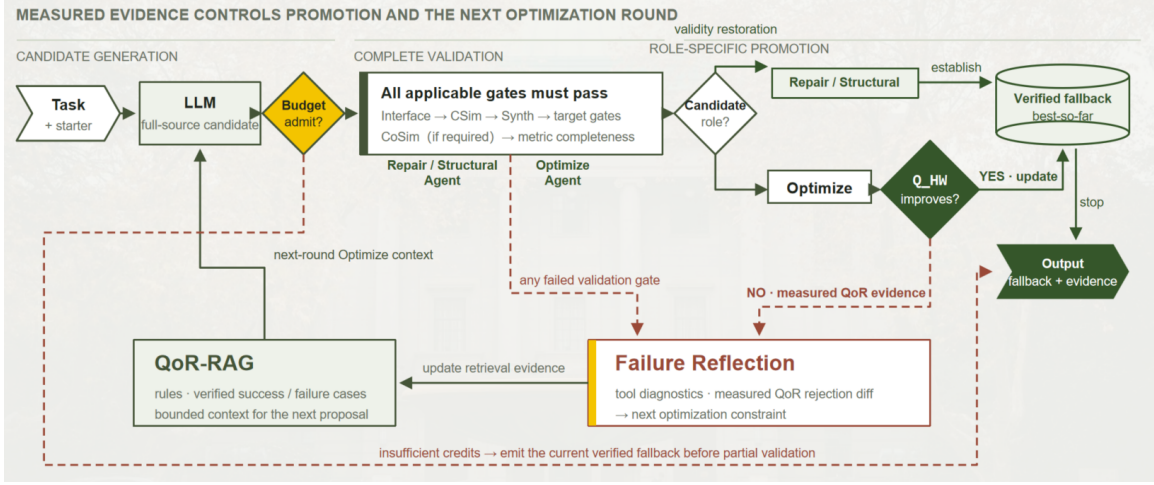


Fig. 1. **The Verified-Candidate Loop.** A candidate replaces the verified fallback only after every applicable gate passes and the composite Q_{HW} improves.

repositories and released with our artifacts [6], with matched task budgets, temperature, token limit, and tool timeouts. Only LLM-API infrastructure errors are rerun; a chronological retry replaces the prior record regardless of its outcome, leaving exactly one final record per task. Completion is the number of completed final records divided by 150. Our separate evaluator reruns every output that passes the public gates using finite held-out CSim, synthesis, and required CoSim gates; these are not the organizer’s final hidden benchmarks. The score columns apply Eq. (1) to the same 24 tasks.

TABLE I
FINAL CAMPAIGN RESULTS. COMPLETION USES THE FIXED 150-TASK DENOMINATOR; HELD-OUT IS CONDITIONAL ON COMPLETION. SCORES USE THE PAIRED SUBSET.

Configuration	Completion	Held-out	Starter	Reference
DeepSeek V4 Pro	143/150	143/143	74.98	75.69
Qwen3.5-122B-A10B	140/150	140/140	77.77	78.93
Qwen3.6-27B	141/150	141/141	77.06	78.56

Results. DeepSeek records the highest completion at 143/150 (95.3%), followed by Qwen3.6 at 141/150 (94.0%) and Qwen3.5 at 140/150 (93.3%). Together, 424/450 final endpoint–task records pass the public gates; all 424 also pass our evaluator-side held-out suite. The remaining 26 records are retained as non-completions in the fixed-denominator rates.

On the 24 tasks scoreable for every configuration, Qwen3.5 records the highest starter- and reference-anchored means. Its reference mean is 0.37 above Qwen3.6. These QoR values use the fixed composite utility defined in Section III.

Token use. Observed all-attempt consumption, including retries, is 2.128M tokens for DeepSeek, ≥ 4.261 M for Qwen3.5, and ≥ 5.398 M for Qwen3.6. Failed Qwen requests omit usage, so both Qwen totals are lower bounds.

Reproduction and limits. Campaigns share the controller, prompts, gates, score, and budgets; endpoint adapters and provider reasoning differ. Thus one campaign per configuration supports descriptive, not causal, comparison. The public

artifact repository [6] provides code, run records, hashes, and aggregation scripts; hosted reruns may vary.

V. CONCLUSION

The VCL preserves a verified fallback under finite validation gates. Across one campaign per configuration, completion ranges from 140/150 to 143/150; all 424 outputs that pass the public gates also pass our evaluator-side held-out suite, which is not the organizer’s final hidden benchmark. Qwen3.5 has the highest paired composite-score means (77.77/78.93), whereas DeepSeek uses the fewest observed tokens.

All configurations complete 25/25 synthesis-repair tasks; DeepSeek, Qwen3.5, and Qwen3.6 respectively lead functional repair (22/25), code generation (25/25), and structural repair (25/25).

These results describe deployed configurations rather than causal model effects. The public repository provides code, run records, and aggregation scripts. Repeated campaigns, component ablations, and score-sensitivity studies remain future work.

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